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Magnetic Reversal Stratigraphy of the Lower Triassic Moenkopi Formation of Western Colorado

ABSTRACT

Oriented cores, taken at stratigraphic intervals of 9 to 12 inches, have been collected from the basal 500 feet of the Moenkopi Formation along the Dolores River in western Colorado. The NRM and the demagnetized data from these samples indicate that the Lower Triassic magnetic field is characterized by at least eleven reversals. Samples from the normal polarity intervals are initially well clustered and remain so after AC or thermal demagnetization. Samples with reversed polarity are initially scattered and remain so after AC demagnetization. However, after thermal demagnetization to 500° C, they become well clustered and have a direction approximately 180° from that of the normal intervals.

All samples belong to either the normal or reversed groups, and no samples transitional between normal and reverse were observed.

A tentative correlation of the resulting polarity reversal sequence with that from the Chugwater Formation of Wyoming is proposed. A magnetic anomaly pattern similar to this polarity reversal sequence is observed in magnetic profiles at sea, indicating that polarity reversal sequences may be a useful means of both regional correlation and calibration of the sea-floor spreading rates postulated for the ocean basins.

INTRODUCTION

For the Late Tertiary and Quaternary, polarity reversal sequences have been established in radiometrically dated lavas by Cox and others (1964). Similar polarity reversal sequences have been observed in deep-sea cores (for example, Opdyke and others, 1966; Watkins and Goodell, 1967). Recent studies of the magnetic anomaly patterns at sea (Vine, 1966; Heirtzler and Hayes, 1967; Heirtzler and others, 1968) have enabled us to develop a magnetic reversal sequence for the entire Tertiary and to extend these anomaly patterns into the Mesozoic and perhaps into the latest Paleozoic. These extensions, however, are based largely on the assumption of constant sea-floor spreading rates and thus may be considerably in error. It is, therefore, essential that polarity reversal sequences of known age be studied in rocks of Mesozoic and Paleozoic age in order to provide a framework for interpretation of the magnetic anomaly patterns observed at sea.

For rocks older than Late Tertiary, few polarity reversal sequences have been published, although reversals are known from all portions of the stratigraphic record with the exception of the long reversed polarity interval in the Permian (Irving, 1964), and perhaps a long normal polarity interval in the Cretaceous (Helsley and Steiner, 1969).

Several attempts have been made to use polarity reversal sequences in stratigraphic correlation, notably by Khramov (1958), Picard (1964), and Burek (1967). These studies have been somewhat inconclusive since the polarity reversal sequences for these time intervals were not established sufficiently clearly prior to using them in stratigraphic correlation efforts. The purpose of this paper is to define the polarity reversal sequence for rocks of Lower Triassic age from the Colorado Plateau. This polarity sequence can then be used as a means of regional and perhaps worldwide correlation of Triassic rocks as well as a means of identification of magnetic anomaly patterns at sea.

ACKNOWLEDGMENTS

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PREVIOUS WORK

Observations made by Runcorn (1956), Kintzinger (1957), and Collinson and Runcorn (1960) have shown that rocks of Lower Triassic age in the western United States characteristically have both normal and reversed polarities, although these authors did not attempt to define the polarity reversal sequence in these rock groups. The Lower Triassic Chugwater Formation of Wyoming has been more thoroughly studied by Picard (1964); there he was able to find 11 magnetic zones (but only 7 polarity zones), some of which could apparently be correlated over large distances.

GEOLOGIC SETTING

The Moenkopi Formation crops out throughout the Colorado Plateau region in Colorado, New Mexico, Arizona, Utah, and Nevada. Its regional stratigraphy and history are discussed by McKee (1954). In the western portions of its outcrop region, the Moenkopi Formation is characterized by a marine sequence including gray shales, carbonates, and evaporites in which red to maroon shales are interbedded. To the east, the formation becomes more clastic, and in eastern Utah and western Colorado, it consists of brown and red-brown siltstone and shale with minor amounts of gypsum and a few locally developed conglomerate lenses.

The region sampled for these studies is in the clastic facies of the Moenkopi Formation south of Gateway, Colorado, (Fig. 1) in an area in which the geology has been described by Shoemaker (1955). Along the Dolores River, the Moenkopi can be subdivided into three members (Shoemaker and Newman, 1959), although a fourth (uppermost) member is present a few miles to the west. Bedding within the Moenkopi Formation is generally even and continuous, which greatly facilitates lithologic correlations. In ascending sequence, the three prominent lithologic members exposed along the Dolores River are: (1) a basal unit of muddy sandstone interbedded with minor amounts of siltstone overlying a thin white to lavender gypsum layer, (2) a unit of well-bedded siltstone interbedded with local conglomerate, and (3) a siltstone unit with interstratified sandstone. The Moenkopi Formation overlies the lavender to purple arkosic sandstone of the Permian Cutler Formation with an angular unconformity of several degrees. To the east, it thins abruptly against the Uncompahgre Uplift. These highlands provided the source for many of the local conglomerate units that are present in this portion of the section. To the east, the Moenkopi Formation in this area is overlain by the Upper Triassic(?) Chinle Formation with up to 7° angular unconformity, but is apparently locally conformable in regions to the west and north (Shoemaker and Newman, 1959).

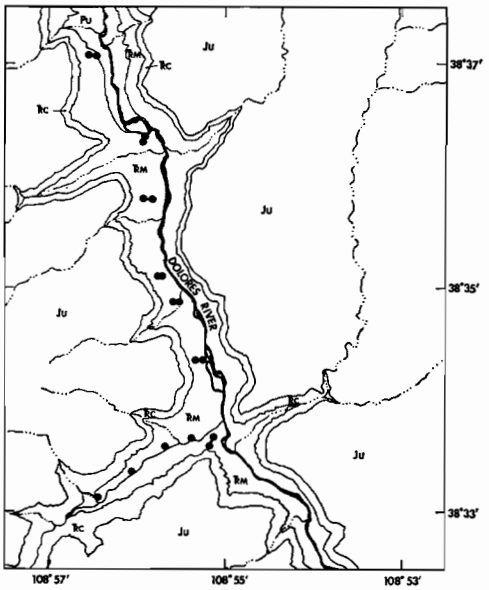


Figure 1. Location map showing the sampling localities (••) within the Moenkopi Formation. Pu = Permian Cutler Formation, Trm = Triassic Moenkopi Formation, Trc = Triassic Chinle Formation, Ju = undifferentiated Jurassic Formations.

The age of the Moenkopi Formation of this region is well established as Lower Triassic (Dane, 1935; Shoemaker and Newman, 1959); however, the range of time represented by the formation is poorly known. On the basis of regional correlation with other Permian and Triassic strata, Shoemaker and Newman (1959) conclude that the lowermost member of the Moenkopi may be either Permian or Triassic and that it is likely that the boundary lies within the lowest member. The uppermost units of the Moenkopi Formation are considered by the same authors to be perhaps as young as Middle Triassic. Thus, one can only

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say that the reversal sequence described below is of Lower Triassic age and that it may represent the reversal sequence from uppermost Permian time to the upper part of the Lower Triassic. No samples were taken from the portion of the section considered to be possibly of Middle Triassic age.

COLLECTION TECHNIQUES

Oriented samples were collected at 9- to 12-inch stratigraphic intervals throughout the lower half of the Moenkopi Formation exposed along the Dolores River, 15 miles south of Gateway, Colorado. All samples were collected by means of a hand-held gasoline powered drill (Helsley, 1967b) and were oriented by means of a magnetic compass. The interval of detailed sampling is a composite of seven sections that were correlated on the basis of prominent lithologic units exposed in the cliffs along the Dolores River.

In the area sampled, a prominant white gypsum layer at the base of the Moenkopi lies directly on the purple arkosic sandstone of the Cutler Formation. Samples were collected from all lithologies except the conglomerates and a 25-foot covered interval approximately 100 feet above the base of the formation. The lower 550 feet of the formation were sampled in detail. Additional reconnaissance samples were taken at 10- to 30-foot intervals in the upper part of the formation, except that no samples were collected from the upper 120 feet of the formation. Since the Moenkopi Formation is approximately 1000 feet thick in this region, the detailed results presented below cover only about half of Moenkopi time.

Samples collected during the initial search for reversals were cut to 1-inch cylinders in the field and measured in the field with a portable air-turbine spinner magnetometer (Helsley, 1967a). Subsequent samples taken in continuous stratigraphic sequence were cut to 1-inch cylinders upon return to the laboratory and were initially measured on the same air-turbine spinner magnetometer. Measurements made during the alternating current and thermal demagnetization experiments, including remeasurement of the initial magnetization, were made on a PAR SM-1 magnetometer.

DISCUSSION OF NRM RESULTS

The natural remanent magnetization (NRM) of all samples from the lower 550 of the formation is summarized in Figure 2. Several features are immediately apparent. The reversed groups have a declination near 150° and a slightly

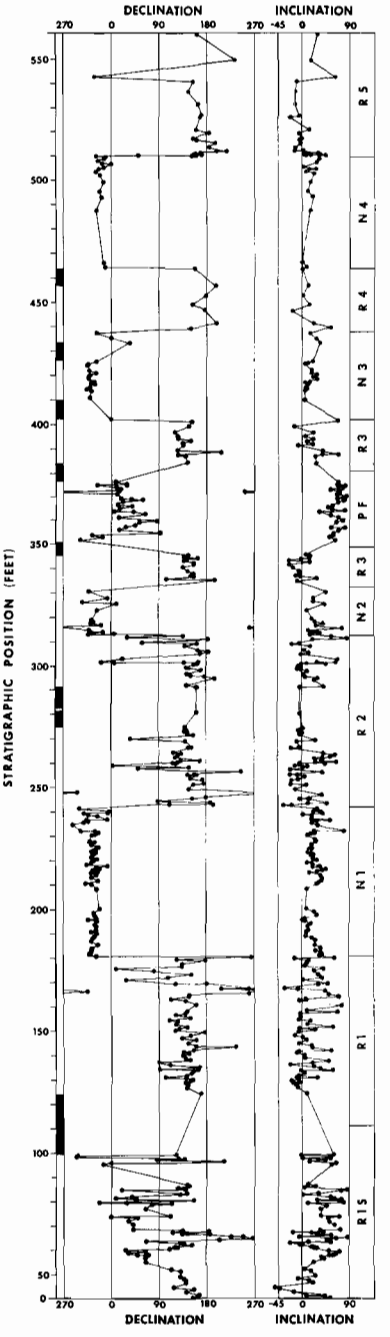


Figure 2. Declination and inclination of the NRM observed in the lower half of the Moenkopi Formation. Black bars at left indicate intervals where no samples were collected due to lack of exposure or presence of a conglomerate. Abbreviations at right indicate polarity zones.

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TABLE 1. NRM OF NORMAL INTERVALS

Interval	Dec	Inc	N	R	K	Alpha	Del	Long.	Lat.	DP	DM
N 1 NRM	344.1	24.0	62	59.88	28.73	3.4	15.0	−103.9	60.3	2.0	3.7
N 2 NRM	343.3	27.8	13	12.39	19.67	9.6	17.6	−107.3	62.0	5.7	10.5
N 3 NRM	339.4	16.2	17	16.70	53.61	4.9	10.7	−107.8	54.4	2.6	5.1
N 4 NRM	352.1	18.6	23	22.31	32.02	5.4	14.0	−86.9	60.1	2.9	5.7
N 5 NRM	337.1	20.0	10	9.86	63.50	6.1	9.7	−113.1	55.1	3.3	6.4
MEAN	343.2	21.4	5	4.97	131.74	6.7	6.3	−104.1	58.7	3.7	7.1

negative (upward) inclination. The normal groups have a declination of about 335° and a positive inclination of about 20°. Thus, the normal and reversed results are approximately, although not precisely, 180° apart. Several portions of the section show evidence of considerable instability. This instability is most noticeable in the lower 60 feet of the section where the results are quite scattered, although they show very prominent systematic changes with stratigraphic position (Fig. 2), suggesting that not all of the scatter is due to secondary components. Another conspicuously anomalous unit occurs between 350 and 375 feet above the base of the section. This unit, bounded by conglomerate lenses, has a fairly consistent orientation with a steep (70°) inclination and a declination of about 35°. At first glance, one might assume that this is an unstable magnetization acquired in the present field, perhaps, during weathering. However, upon closer examination, none of these samples are near the axial dipolar field, and only a few are near the field observed today. A concentration of points with a similar direction is present in the scattered results from the lower 100 feet of the Moenkopi section.

Perhaps the most striking feature in Figure 2 is the marked difference in the degree of scatter of the results from the reversed and normal groups. The portions of the section that have normal polarity are generally quite tightly grouped and have dispersions characteristic of many red bed sequences; the zones of reversed polarity, however, are much more scattered, although by no means random (Table 2 and Irving, 1964, p. 63). This scatter is not in the form of a streak as was observed in the Chinle Formation by Collinson and Runcorn (1960). Instead, it is a scatter in both inclination and declination much larger than expected from the addition of a more or less random vector parallel to the present earth's field with that of the typical reversed Triassic direction. Nevertheless, no other cause for this large dispersion has been found.

The Schmidt equal area projection plots of the NRM from the 5 normal intervals and a summary of their mean directions are shown in Figure 3. Their statistical parameters (Fisher, 1953; Watson, 1956) are summarized in Table 1. All of these groups are well clustered and clearly different from the present earth's field. Moreover, they show no obvious evidence of streaking toward the present earth's field and give results that are very similar to other Triassic results from the Colorado Plateau region (Irving, 1964). Thus, they have been considered to be stable and to represent a true direction for the Triassic period. This assumption is substantiated by the AC and thermal demagnetization experiments discussed below.

The NRM directions of the reversed groups are presented in Figure 4, and their statistical parameters are given in Table 2. The assignment of points to the reversed groups versus the normal groups has been made on the basis of the prominent declination and inclination

TABLE 2. NRM OF REVERSED INTERVALS (ALL SAMPLES)

Interval	Dec	Inc	N	R	K	Alpha	Del	Long.	Lat.	DP	DM
R 1 NRM ALL	148.2	34.8	136	94.61	3.26	8.0	44.7	75.6	−25.0	5.3	9.2
R 2 NRM ALL	165.8	6.9	45	38.96	7.29	8.5	29.7	88.2	−45.9	4.3	8.5
R 3 NRM ALL	166.4	5.2	34	31.39	12.65	7.2	22.4	88.8	−46.9	3.6	7.2
R 4 NRM ALL	177.9	13.3	7	6.29	8.47	22.0	25.8	106.0	−44.6	11.5	22.5
R 5 NRM ALL	174.7	6.4	21	18.81	9.12	11.1	26.2	101.1	−47.9	5.6	11.2
R 6 NRM ALL	154.6	3.0	25	22.06	8.15	10 8	27.8	72.7	−43.6	5.4	10.8
MEAN	165.0	11.7	6	5.80	25.64	13.5	14.6	88.2	−43.4	7.0	13.7

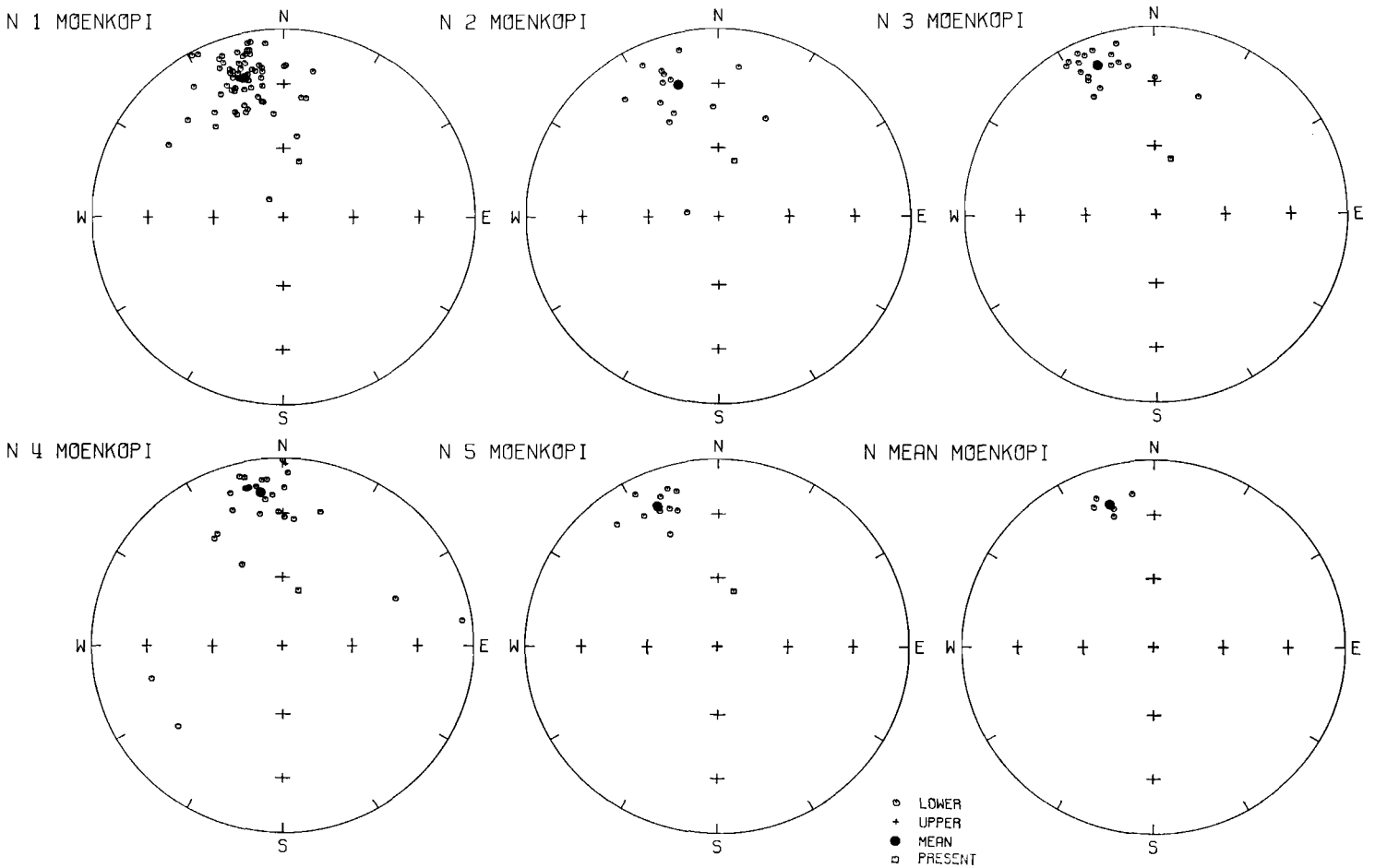


Figure 3. Equal area plots of the NRM directions for the normal intervals and summary plot of the means of all the normal intervals. "Present" is present field direction.

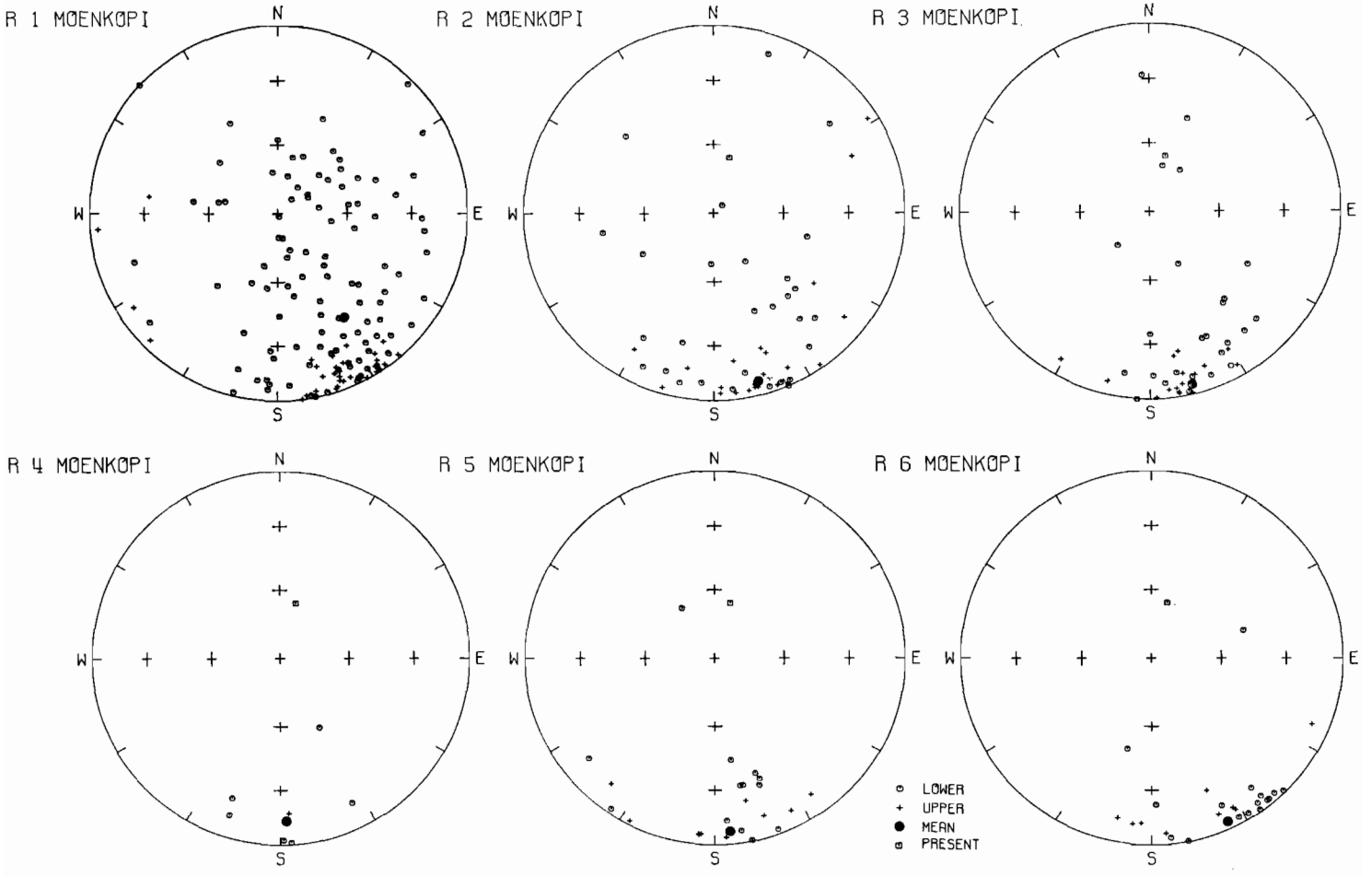


Figure 4. Equal area plots of the NRM directions for the reversed intervals. "Present" is present field direction.

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changes observable in the stratigraphic plot shown in Figure 2. The marked difference between the reversed and normal groups mentioned above is perhaps even more prominent when Figure 4 is compared with Figure 3. Nevertheless, within each of these groups, except Figure 4 (R_4), a fairly tight grouping exists between 150° and 180° declination and with a near zero or slightly negative inclination. This group of points is approximately antiparallel to the well-clustered normal groupings shown in Figure 3. Even though there is a very large scatter in the reversed results, very few observations lie within 30° of the present field direction. The NRM of the reversed groups can therefore be summarized as being more scattered than that for the normal intervals with the clustered portion of the NRM's being approximately 160° from the normal mean directions with a very slight, but not at all prominent, streak toward the present earth's field.

As stated previously, two stratigraphic intervals (R_1S and PF) have results that are very different from either the normal or reversed groups. These are shown in Figure 5. In Figure 5(a), the results from the stratigraphic interval 350 to 375 feet above the base of the formation (interval PF) are seen to be relatively well grouped, near the present field, but still significantly removed from it, and very different from the typical normal or reversed groups. The other group (R_1S), shown in Figure 5(b), comes from the lowest 100 feet of the formation. It seems to consist of two separate groups, one of which is a very scattered reversed direction with a suggestion of streaking toward the present earth's field, while the other is a prominent group that plots to the east and somewhat south of the present earth's field and is very similar to the direction observed for the samples shown in Figure 5(a). However, if one compares the stratigraphic plot shown in Figure 2 with the direction plot shown in Figure 5(b) there seems to be little consistency in direction with stratigraphic position within this interval in contrast to that for the interval shown in Figure 5(a). Sequential samples in the interval 40 to 70 feet above the base of the Moenkopi Formation show systematic changes away from the reversed Triassic field direction toward the present field, indicating that some of the scatter is caused by the acquisition of a secondary component in the present earth's field. Both of the groups may well be unstable and consist of mixtures of a Triassic field direction with a field similar to that of the present day. For the scattered samples shown in Figure 5(b), this assumption is supported by the demagnetization results discussed below. The directions shown in Figure 5(a), however, do not change during demagnetization and thus may represent the record of an anomalous field

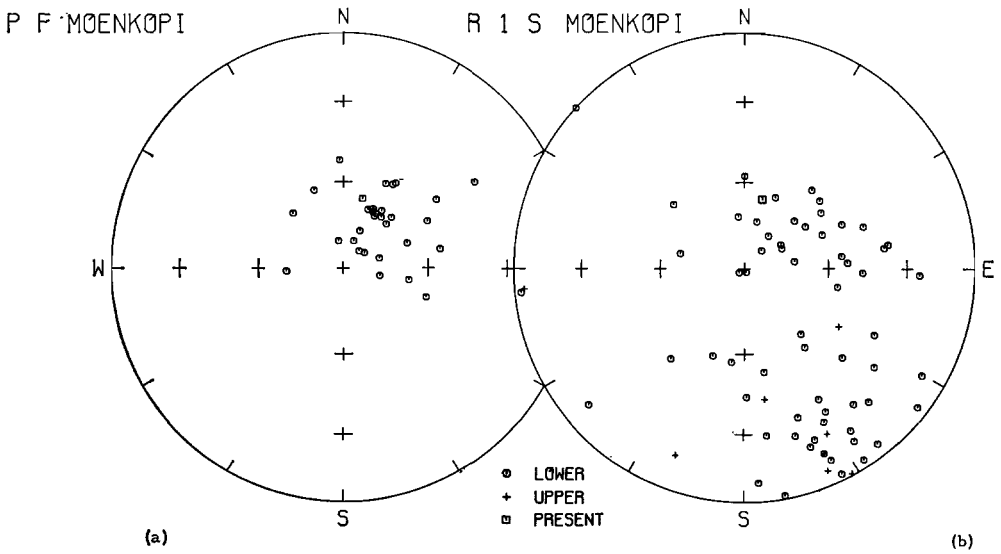


Figure 5. Plots of the NRM directions: (a) the anomalous interval near the present field direction, (b) the widely scattered results from the lower portion of the first reversed interval.

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direction similar to the anomalous direction observed in rocks of Permian age from the Dunkard Series (Helsley, 1965, Fig. 3b) or that found for Eocene rocks by Cox (1957).

DEMAGNETIZATION EXPERIMENTS

The magnetic stability of the samples from the Moenkopi Formation has been studied by both alternating current and thermal methods. The alternating current demagnetization experiments were made on 420 cycle equipment similar to that described by Patton and Fitch (1962). All samples were tumbled about three axes during the AC demagnetization process in a region in which the residual D.C. field was less than 200 gammas. The applied AC field was varied smoothly from its maximum value to zero by precisely controlling the excitation current supplied to the 420 cycle motor-generator. No systematic spurious components were observed even at field values of up to 2800 oersted.

Thermal demagnetization was accomplished in a conventional non-inductively wound electrical furnace, the samples being cooled in a region where the residual field was less than ± 10 gammas. Temperature was monitored at each level of the furnace by means of platinum thermocouples that were referred to an ice bath and read on a digital voltmeter.

Initial progressive AC demagnetization studies were made on about 10 percent of the samples with applied fields ranging from 44 to 1400 oersteds. This step-wise demagnetization showed that the magnetization was stable, that is, its intensity did not change rapidly with applied field, and that the coercive forces of the true NRM and any perturbing secondary components were approximately the same. Consequently, little decrease in scatter was accomplished as a result of the AC demagnetization experiments, as can be seen in the three sets of results given in Table 3.

Since AC demagnetization had little effect on the observed directions of the NRM, progressive thermal demagnetization experiments were made on another set of samples selected primarily from the scattered portions of the reversed groups. These experiments produced large changes in the direction of magnetization, but only a small effect on the observed intensity until temperatures in excess of 600° C were reached. Figure 6 shows the response of a few of the samples to the progressive heating experiments.

The normal samples changed very little for temperatures below 600° C, their directions generally changing by only a few degrees, and their intensities decreasing from 10 to 20 percent at temperatures below 600° C. The reversed and scattered samples, on the other hand, underwent large changes in direction (in some cases more than 90°; for example, samples 147 and 132, Fig. 6) accompanied by a change of 10 to 50 percent in intensity. Above 600° C, and particularly above 650° C, the intensity of both normal and reversed samples decreased very rapidly. These rapid decreases in intensity were commonly associated with rather large and more or less random changes in direction, suggesting that a significant secondary component may have been acquired during the cooling of the samples from temperatures in excess of 650° C.

The demagnetization curves presented in Figure 6 are from short intervals within the first reversed and first normal intervals and, although their general behavior is similar to that for all of the other samples studied, their intensity differences are much more pronounced. As is readily seen in Figure 6, all of the normal samples have a much larger NRM than do the reversed samples (by a factor of about 8). This might be explained by assuming that all samples had acquired a large secondary component parallel to the present field. Since

TABLE 3. RESULTS OF AC DEMAGNETIZATION

Interval	Dec	Inc	N	R	K	Alpha	Del	Long.	Lat.	DP	DM
R1S 114–140 ORIG	140.5	43.2	32	23.14	3.50	16.0	42.6	72.0	−16.3	12.3	19.9
R1S 114–140 AC175	148.4	33.9	32	24.06	3.90	14.9	40.4	75.4	−25.6	9.7	17.0
R1S 114–140 AC350	155.4	28.0	32	25.17	4.50	13.5	37.4	80.6	−31.7	8.1	14.7
R 1 291–313 ORIG	161.6	3.7	20	18.08	9.88	10.9	25.1	81.8	−46.1	5.5	10.9
R 1 291–313 AC 44	167.0	1.8	20	18.12	10.12	10.8	24.8	88.9	−48.7	5.4	10.8
R 1 291–313 AC175	164.5	−0.9	20	18.30	11.16	10.2	23.6	84.7	−49.3	5.1	10.2
R 1 291–313 AC350	168.2	2.6	20	18.29	11.09	10.3	23.7	90.9	−48.7	5.1	10.3
R 3 432–445 ORIG	144.6	14.1	14	10.94	4.25	22.0	37.9	65.2	−33.7	11.5	22.5
R 3 432–445 AC175	145.8	10.1	14	10.66	3.90	23.2	40.0	65.1	−36.0	11.9	23.5

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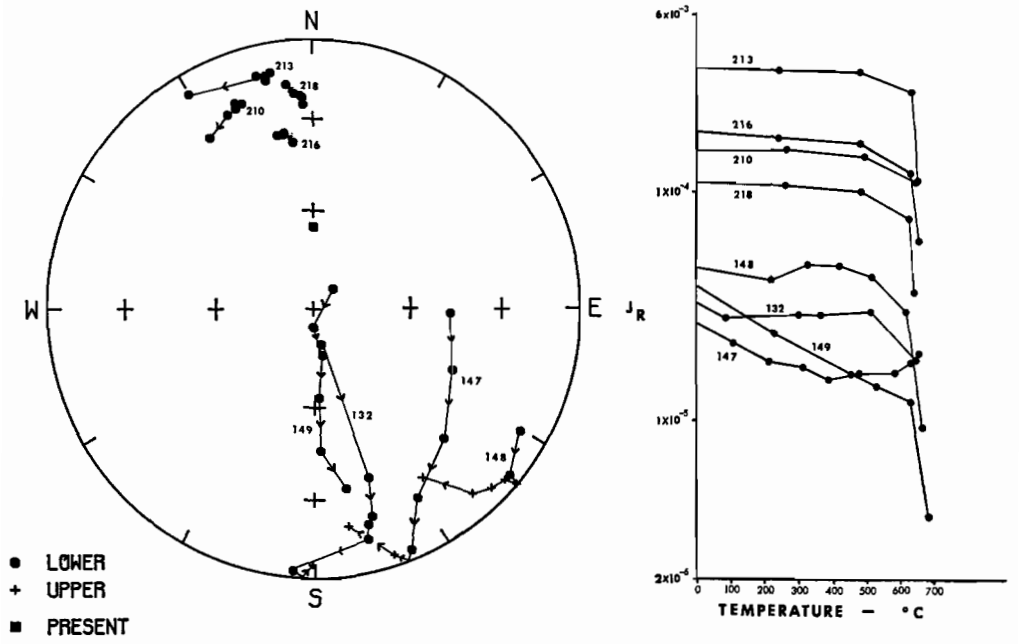


Figure 6. Response of selected samples to progressive thermal demagnetization: (a) changes in direction, (b) changes in intensity. Arrows indicate the direction of change with increasing temperature. Intensities are for 12 cc sample. "Present" is the axial dipolar field direction.

this would be approximately opposite to the reversed NRM direction, a lower intensity would be expected for the reversed samples and a larger value for the normal group, which is exactly what is observed. This hypothesis could easily account for differences of a factor of 3 in the initial NRM data and could possibly be stretched to the factor of 8 observed.

Thermal demagnetization, however, should remove the secondary components from the normal as well as the reversed samples, and thus the ratio of their remaining magnetizations should approach unity as demagnetization progresses. This is not observed. If any trend is present, it is the opposite; namely, the ratio of remaining magnetization (normal/reverse) becomes greater. This suggests that the intensity of the reversed samples has an origin that is in some way very different from that of the normal group.

Since thermal demagnetization seemed to be effective in removing secondary components from the scattered reversed data, it was decided to treat all samples by thermal methods. However, after thermally demagnetizing the first normal interval, the changes in direction and intensity for the already well-grouped normal data were so small that additional thermal demagnetization of normal samples did not seem worth while. The changes in direction that occurred at 500° C for a portion of the N_1 samples are shown in Figure 7 and summarized in Table 4.

Many samples could not be thermally demagnetized to 500° C, for a number of the specimens had broken during drilling in the field, and it was necessary to glue them together with epoxy or Duco cement, both of which break down at relatively low temperatures. Thus only those samples that were unbroken or had only small amounts of glue were used in the demagnetization experiments. It is believed that no undue bias was introduced in this selection since the degree of

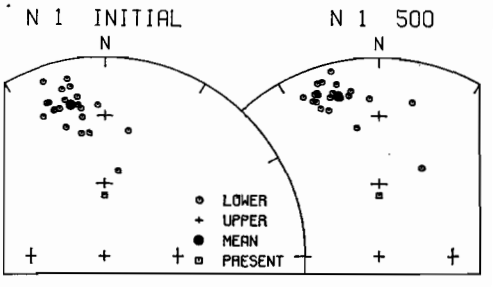


Figure 7. Response of selected samples from the N_1 interval to 500° C thermal demagnetization.

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TABLE 4. DATA FOR INTERVAL N1 BEFORE AND AFTER THERMAL DEMAGNETIZATION

Interval	Dec	Inc	N	R	K	Alpha	Del	Long.	Lat.	DP	DM
N 1 INITIAL	347.2	24.5	22	21.45	38.26	5.1	12.8	−98.4	61.8	2.9	5.4
N 1 500 C	345.8	18.0	22	21.39	34.41	5.4	13.5	−98.3	58.0	2.9	5.6

breakage had no systematic relation to degree of scatter. A temperature of 500° C was chosen as an ideal temperature, for most samples had approached the end of their systematic movement before this temperature was reached, and yet no large spurious displacements had occurred. The results of this thermal demagnetization of the reversed data can be seen in Figure 8.

Even after thermal demagnetization at 500° C, some samples did not approach the main group, as can be seen for all intervals shown in Figure 8. Many of these more scattered samples were subsequently heated to 600° C or 650° C, with little or no systematic change. Thus, the final results presented in Table 5 are in two parts; the first (Table 5A) includes all of the demagnetized results for the various intervals while the data presented in Table 5B has been selected to remove any points lying more than 30° from the mean direction derived from all of the data. These selected data have been used to determine the mean reversed pole position given in Figure 10(c) and Table 7.

One group of anomalous samples did not change upon either AC or thermal demagnetization. These are the samples from the interval between 350 and 375 feet from the base of the formation that were initially well grouped near the present field [NRM shown in Fig. 5(a)]. The NRM of the samples selected for thermal demagnetization are shown in Figure 9(a). These directions are a bit more tightly grouped than are the thermally demagnetized results [Fig. 9(b)], however, no significant change in direction has occurred (Table 6).

The samples in this group come from a stratigraphic interval of 25 feet bounded by conglomerate units 6 to 10 feet thick (*see* Fig. 2), and all samples within this interval give anomalous results. No attempt has been made to sample the identical interval at other localities to see if the anomalous magnetization is persistent throughout the region or if it is due to some unknown local effect. The gross lithology and variations in lithology in this interval appear to be identical with those in the adjacent intervals that give typical reversed Triassic directions. The average intensity of magnetization of these anomalous samples is 4.14×10^{-6} emu/cc, while that of the reversed group immediately below is 1.16×10^{-5} emu/cc, and that of the reversed group above the anomalous interval is 2.53×10^{-6}. These intensities are similar to the means for all reversed samples (3.95×10^{-6} emu/cc) and all normal samples (5.84×10^{-6}). Thus, no marked difference in intensity appears to exist, although there is a large change in intensity with respect to the reversed group immediately below the anomalous interval. One is therefore forced to conclude that the magnetization of this interval is apparently stable and differs from that for the rest of the section only in direction. Moreover, it is tempting to conclude that the magnetization was acquired during a period of time when the Triassic field had an anomalous configuration. Interestingly enough, the pole derived from these observations is in the Atlantic Ocean off Newfoundland not far from the positions reported by Grommé and Gluskoter (1965) for the Franciscan Formation (Cretaceous) and by Cox (1957) for the Siletz River Volcanics (Eocene).

Even after thermal demagnetization, the lower half of the first reversed interval (R_1S) is much more scattered than any of the other intervals. The upper portion of the first reversed interval (R_1) initially has a scatter similar to the other reversed intervals and clusters well upon thermal demagnetization (Fig. 8). Within the R_1S interval, two groups of samples seem to be present [*see* Fig. 5(b) and earlier discussion]: one about 30° from the horizontal in the southeast quadrant, and one about 30° from the vertical in the northeast quadrant. Unfortunately, most of the samples from the northeast group were very friable or had been broken during sampling, and thus only a few could be thermally demagnetized.

Upon thermal demagnetization, the selected R_1S samples from the southeast group move to a cluster on the upper hemisphere just as did those from the upper portions of the first reversed interval [*compare* Fig. 4 (R_1) *with* Fig. 8 (R_1)].

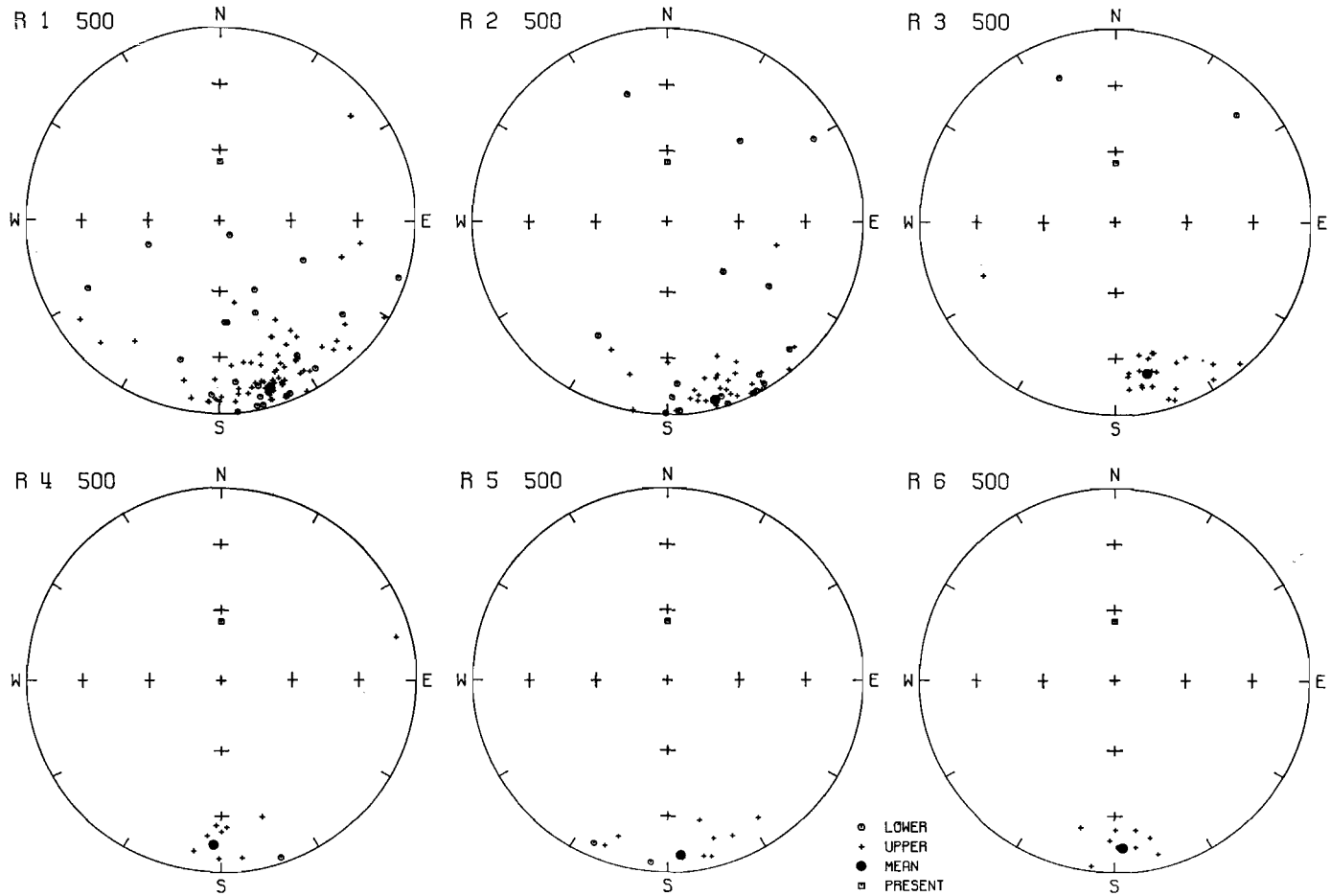


Figure 8. Directions of magnetization for the reversed intervals after 500° C thermal demagnetization. "Present" is geocentric dipolar field.

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The few samples from the northeast group, that is, those that have the same anomalous NRM direction as do those from apparently stable, yet anomalous, interval (PF) shown in Figure 9, change on thermal demagnetization and have progressive thermal demagnetization paths that move toward the reversed Triassic direction given by the other R_1 and R_1S samples. However, at temperatures up to 650° C, many of these samples have not moved all the way to the Triassic directions given by the adjacent samples (for example, sample 149

TABLE 5. A. REVERSED INTERVALS AFTER 500 C THERMAL DEMAGNETIZATION

Interval	Dec	Inc	N	R	K	Alpha	Del	Long.	Lat.	DP	DM
RIS 500 C	165.2	−3.9	44	37.37	6.48	9.2	31.5	85.1	−50.9	4.6	9.2
R 1 500 C	163.5	−14.8	43	38.91	10.28	7.1	25.0	79.1	−55.6	3.8	7.3
R 2 500 C	164.6	−6.2	46	40.59	8.32	7.8	27.8	83.5	−51.8	3.9	7.8
R 3 500 C	167.8	−19.6	24	22.25	13.12	8.5	21.9	84.8	−59.5	4.6	8.9
R 4 500 C	183.4	−14.6	10	8.97	8.75	17.3	26.0	115.4	−58.7	9.1	17.7
R 5 500 C	175.8	−10.0	11	10.36	15.54	12.0	19.6	101.3	−56.2	6.1	12.1
R 6 500 C	177.4	−12.9	9	8.73	29.97	9.6	14.0	104.0	−57.8	5.0	9.7
Mean	171.1	−11.8	7	6.92	76.16	7.0	8.6	92.7	−56.4	3.6	7.1

B. REVERSED INTERVALS AFTER 500 C THERMAL DEMAGNETIZATION (SCATTERED DATA REMOVED)

Interval	Dec	Inc	N	R	K	Alpha	Del	Long.	Lat.	DP	DM
R 1 500 C	162.9	−9.8	93	79.49	6.81	6.1	30.9	79.8	−52.9	3.1	6.2
R 2 500 C	164.6	−6.2	46	40.59	8.32	7.8	27.8	83.5	−51.8	3.9	7.8
R 3 500 C	167.8	−19.6	24	22.25	13.12	8.5	21.9	84.8	−59.5	4.6	8.9
R 4 500 C	183.4	−14.6	10	8.97	8.75	17.3	26.0	115.4	−58.7	9.1	17.7
R 5 500 C	175.8	−10.0	11	10.36	15.54	12.0	19.6	101.3	−56.2	6.1	12.1
R 6 500 C	177.4	−12.9	9	8.73	29.97	9.6	14.0	104.0	−57.8	5.0	9.7
Mean	172.0	−12.3	6	5.94	78.05	7.6	8.4	94.2	−56.8	4.0	7.8

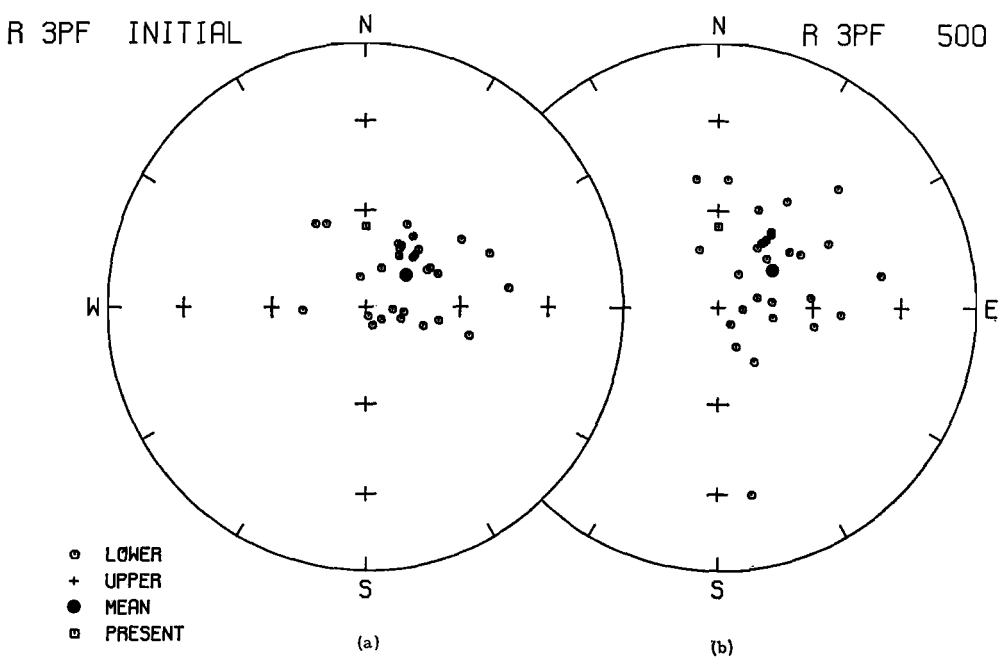


Figure 9. NRM directions before (a) and after (b) 500° C thermal demagnetization for the anomalous (PF) interval within the R_3 reversed interval.

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TABLE 6. PF GROUP BEFORE AND AFTER 500 C THERMAL DEMAGNETIZATION

Interval	Dec	Inc	N	R	K	Alpha	Del	Long.	Lat.	DP	DM
PF INITIAL	52.3	73.8	30	28.59	20.58	5.9	17.6	69.6	51.2	9.6	10.7
PF 500 C	56.9	68.7	30	27.64	12.29	7.8	22.7	57.2	49.0	11.2	13.2

in Fig. 6), and thus they account for many of the more scattered points in Figure 8 (R_1). These results indicate that the northeast cluster belong to the reversed group but also suggest that part of their secondary NRM is "hard" and perhaps was acquired by chemical weathering in the modern environment. This may also be the origin of the magnetization observed in the anomalous interval shown in Figure 9, but if so, no evidence of the "original" magnetization could be found by either thermal or AC demagnetization techniques.

POLE POSITIONS

The pole positions derived from both the NRM and thermal demagnetization measurements presented above in Figures 3 to 9 are given in the Tables 1 through 6, and the means of the normal and reversed intervals, as well as the mean of all of the intervals, are listed in Table 7. Of the normal groups, only interval N_1 was thermally demagnetized. The pole, after thermal demagnetization, differed from the NRM pole by less than 4° (*see* Table 4) and was moved to a shallower angle away from the present field, suggesting that a small amount of secondary magnetization parallel to the present field may have been present. Since this was the most scattered of the normal groups, and the change was so small, no further thermal demagnetization was done on the normal groups. Thus, the NRM directions have been used for the calculation of all normal pole positions and the mean given in Table 7.

The reversed directions underwent large changes, and thus the only reliable pole positions are those for the thermally demagnetized samples. Since some samples did not respond to thermal treatment as well as others, poles were calculated for a selected group of reversed samples (Table 5B), as well as for all the demagnetized samples (Table 5A). It is felt that this selected group of samples provides a more realistic set of reversed poles, for it has eliminated many of the samples that still retained some secondary magnetization. The poles for the NRM data, the thermally demagnetized data, and the selected thermally demagnetized data are shown in Figure 10(a), (b), and (c), respectively. This sequence of poles shows a progressive movement away from the present dipolar field and a marked decrease in scatter as a result of the demagnetization.

The selected demagnetized data from the reversed intervals and the NRM data from the normal intervals are considered to give the best estimates of the pole positions for the Moenkopi Formation. Both sets of poles are well grouped, have cones of confidence at the 95 percent level of about 6.5°, and are about 174° apart. The mean of the normal and reversed groups combined is given by the open circle on Figure 11. The statistical parameters of the normal, reversed, and combined intervals are summarized in Table 7.

If all of the normal intervals had behavior similar to the N_1 group upon being thermally demagnetized, the mean direction would have moved about 4° away from the present pole and the mean of the normal and reversed groups would be changed by less than 2°, which is well within the α_{95} cone of confidence. Using the test of equivalence given by Watson (1956, the data in Table 7 gives $F_{2,2(n-1)} \cong 4.44$, which is greater than the statistic for equivalence 3.49; therefore, the normal and reversed populations are significantly different at the 95 percent level but not at the 99 percent level (statistic for equivalence 5.85).

TABLE 7. POLE DATA FOR N, R, AND N + R INTERVALS

Interval	Dec	Inc	N	R	K	Alpha	Del	Long.	Lat.	DP	DM
N NRM MEAN	343.2	21.4	5	4.97	131.74	6.7	6.3	−104.1	58.7	3.7	7.1
R 500 C SELECTED MEAN	167.4	−13.1	7	6.94	95.93	6.2	7.7	86.0	−56.1	3.2	6.3
MOENKOPI MEAN	345.7	16.6	12	11.87	84.57	4.7	8.4	−98.0	57.2	2.5	4.9

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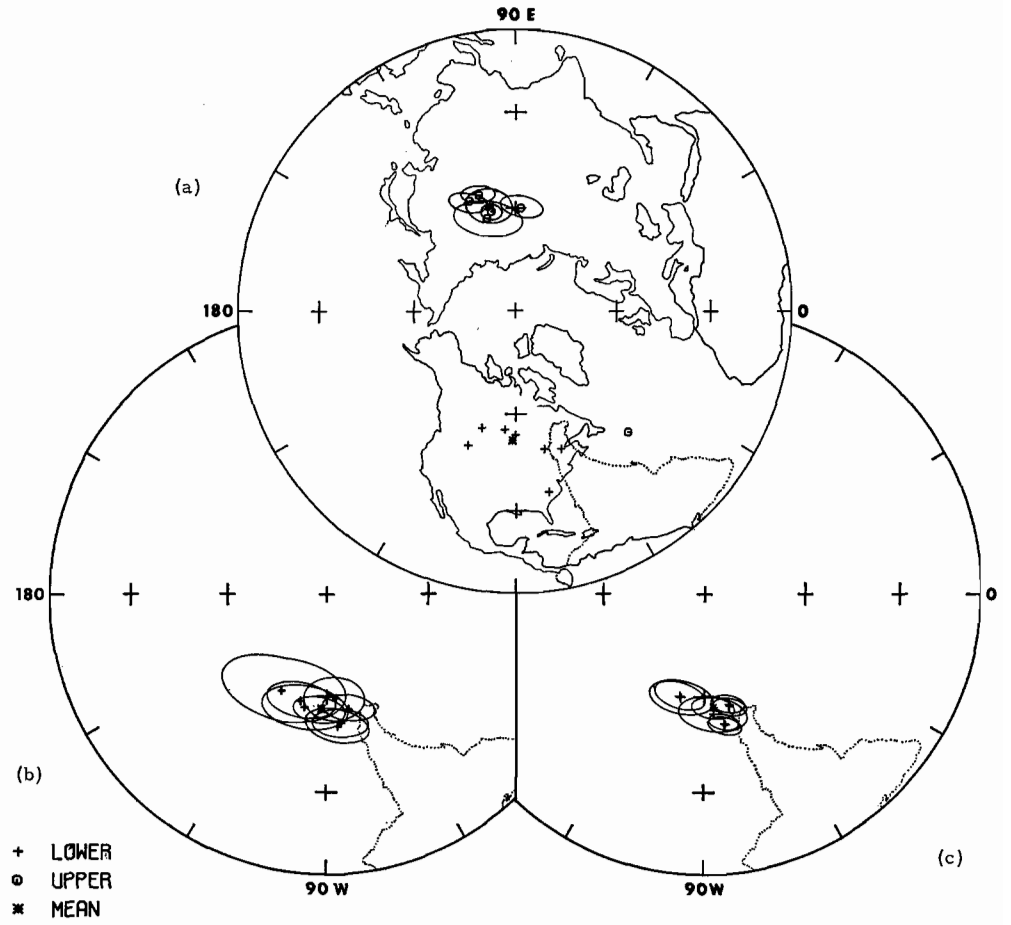


Figure 10. Three plots showing the poles derived from the individual polarity intervals. (a) Pole positions and α_{95} confidence limits for the normal intervals and pole positions for the clustered portions of the reversed data; (b) poles and α_{95} confidence limits for the thermally demagnetized data; (c) poles and α_{95} confidence limits for the reversed intervals after removal of those samples that did not cluster upon thermal demagnetization. Poles for reversed intervals plotted on lower (southern) hemisphere. Dotted outline of South America is on the lower hemisphere.

COMPARISON WITH OTHER TRIASSIC POLES FOR NORTH AMERICA

A summary of the Triassic pole positions for North America is given in Figure 11 and Table 8. The results of this study lie in the center of the main group of poles reported for other Triassic units (*see* Table 8 for references). The only pole position significantly removed from this main group is that of the New Oxford Formation (Graham, 1955) as reported by Irving (1964).

INTENSITY OF MAGNETIZATION

Picard (1964) analyzed all of his intensity data from the Triassic Chugwater Formation in terms of lithologic types and in terms of normal and reversed intervals. He found that the distributions within normal and reversed groups were identical and that the only differences present were between lithologic types. His mean intensity for all samples was 6.5×10^{-6} emu/cc, and his range was from 0.6 to 70.4×10^{-6} emu/cc.

Within the Moenkopi Formation, no rigorous attempt has been made to relate intensity of magnetization to lithologic type. The mean intensity of all samples is 4.56×10^{-6} emu/cc, and the range of intensities is from 0.31 to 53.2×10^{-6} emu/cc. Figure 12 presents histograms of the observed intensities for all normal samples, all reversed samples, all normal

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samples after heating to 500° C, and all reversed samples after heating to 500° C. Surprisingly, there are significant differences between the normal and reversed groups, but no differences between the NRM and thermally demagnetized intensities. The immediate inference is that this difference in intensity of magnetization is due to either a stronger field during normal intervals or to more magnetic material being present in the normal samples. The latter is unlikely, since the bulk susceptibilities of both groups are identical (Fig. 13), indicating that a more or less similar amount of magnetic material is present. Furthermore, it is unlikely that the distribution of magnetic material would be related to polarity. Thus, one is tempted to conclude that the field in which the rocks were magnetized was weaker during reversed intervals than during normal ones.

This conclusion may not be correct, since the intensity distributions are not normal, and a suggestion of a bimodal distribution is present. Variations in lithology could produce a bimodal distribution and, if the lithologies were not uniformly distributed between normal and reversed intervals, significant differences could occur. Therefore, average intensities were found for each polarity interval and are shown in Figure 14. Two features are immediately apparent. First, there is a seemingly systematic increase in intensity upward from the base of the formation with a large amount of variation present in the middle portion (in the vicinity of the local conglomerate lenses). Second, the first reversed interval (which includes about one-fourth of all of the samples) has a significantly lower intensity than the other reversed intervals except for the few samples from the upper portion of R_3. Since the boundary between the lower two members of the Moenkopi lies within the R_2 interval, the observed intensity differences are probably the result of the muddy sandstones of the lower member not being able to acquire as strong a magnetization (be it of chemical or detrital origin) as could the overlying shales and siltstones. A similar difference with lithologic type has been noted by Picard (1964).

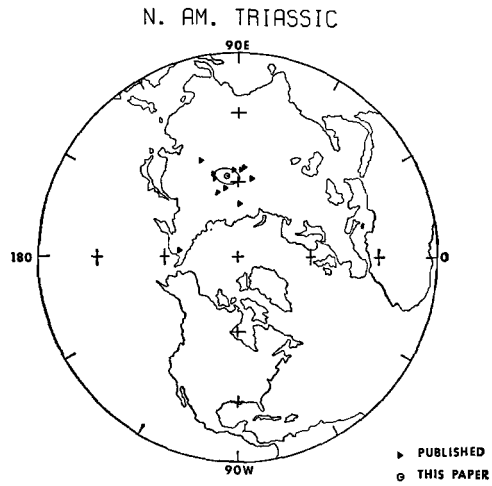


Figure 11. Summary of all Triassic poles for North America (*see* Table 9 for references) including the mean pole for all polarity intervals of the Moenkopi Formation with its associated α_{95} confidence limit (each polarity interval assigned equal weight).

Ignoring the differences between the normal and reversed groups, the mean intensity of the Moenkopi Formation (4.56×10^{-6} emu/cc) is similar but somewhat less than that from the

TABLE 8. SUMMARY OF TRIASSIC POLES

Formation	Lat. N.	Long. E.	D	I	k	α_{95}	Reference*
Connecticut Lavas	58	80	15	27	57	9	2, 4
Chinle FM Combined	55	93	..	..	..	..	7(6, 3)
New Oxford FM	66	174	334	48	36	7	7(6)
Newark Group	63	108	359	25	49	4	7(5, 3, 10)
Connecticut Valley Rocks	54	86	12	14	..	15	7(5)
Chugwater FM	48	112	334	17	51	5	7(3)
Moenkopi FM	57	107	338	19	21	13	7(11, 3, 8)
Springdale SS	55	107	338	16	..	..	7(11)
Massachusetts Lavas	55	88	10	16	41	10	7(5)
Newark Group Diabase Intrusion	62	101	360	23	102	3	1
Nova Scotia Dike	69	98	5	54	193	18	9
Moenkopi FM	57	98	346	17	85	5	This study

*1 = Beck (1965), 2 = Bowker (1960), 3 = Collinson and Runcorn (1960), 4 = De Boer (1968), 5 = Dubois and others (1957), 6 = Graham (1955), 7 = Irving (1964), 8 = Kintzinger (1957), 9 = Larochelle and Wanless (1966), 10 = Opdyke (1961), 11 = Runcorn (1956)

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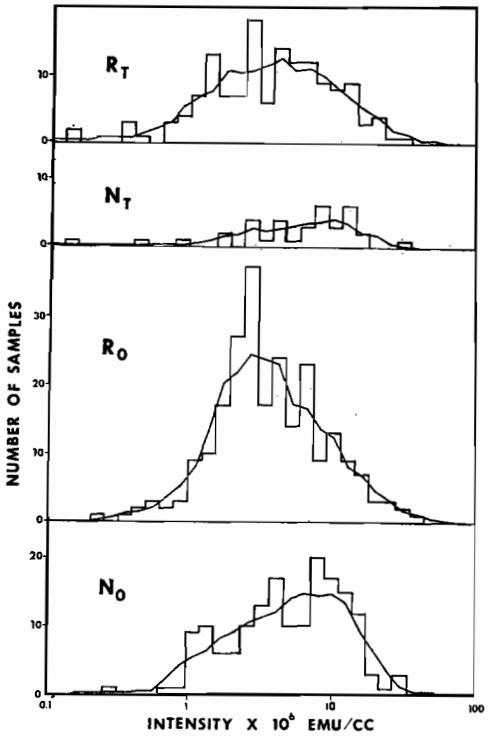


Figure 12. Histograms of the intensity of magnetization within the Moenkopi Formation. Smoothed curves are sliding averages over five intervals.

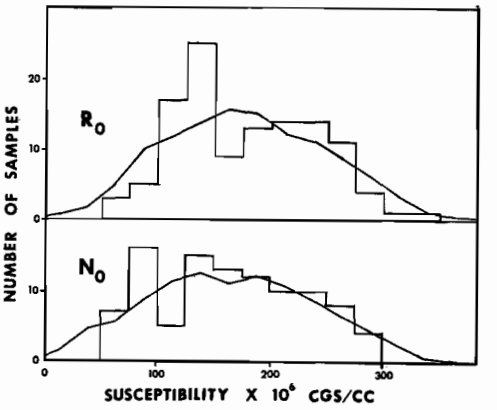


Figure 13. Histograms of the bulk susceptibility of the samples from the Moenkopi Formation. Smoothed curves are sliding averages over 5 intervals.

Chugwater Formation of Wyoming, as reported by Picard in 1964 (6.5×10^{-6} emu/cc).

TRANSITION INTERVALS

One of the initial objectives of this investigation was to study the nature of polarity transition intervals in red sediments. Similar studies have been reported by Khramov (1958), Helsley (1963), and Picard (1964) for other red bed sequences and indicate that these transitions take place in 1 to 4 feet of section. Initial measurements of NRM made in the field readily located two of the transition zones (R_3 to PF and N_4 to R_5); detailed studies were attempted for these intervals. The transition between R_3 and PF proved to be impossible to sample adequately, since the lithology in this interval was conglomeratic sandstone with rare siltstone lenses.

The transition between N_4 and R_5 was much more amenable to detailed sampling, for it was in an interbedded platy siltstone and massive siltstone sequence. After two samplings, the transition zone had been located to about 15 cm and overlapping samples were drilled across the transition. The NRM of these samples is shown in Figure 15(a). Only one sample shows an intermediate direction, and this sample probably is half normal and half reversed, since the adjacent samples are fully normal and reversed, respectively. Thus, the normal to reverse transition takes place in less than 2.5 cm and perhaps in less than 1 cm. This same reversal has been sampled in the same manner about 100 m away, with identical results [Fig. 15(b)].

Eleven other transitions have also been sampled during this study. Density of sampling was about 30 cm (varying from 10 to 40 cm), and no evidence was found for any intermediate directions. This is exactly opposite to the results reported by Picard (1964) from the Chugwater Formation, where several of the transitions took place over 2 or more feet of section; this suggests that the origin of the magnetization of the Moenkopi may be related to chemical diagensis rather than to depositional or detrital effects. The transition shown in Figure 15 lends support to this argument for the reversal takes place abruptly within a virtually homogeneous sandy siltstone layer about 30 cm thick that was probably deposited more or less continuously during a relatively short period of time. Using the same type of argument as that presented by Picard (1964), namely, thickness divided by an estimate of the number of years represented by the formation, the estimated depositional rate for the Moenkopi is about 3 cm per 1000 years. Thus, the N_4 to R_5 transition would have to take place in a few hundred years if the magnetization were of detrital origin. Since this seems

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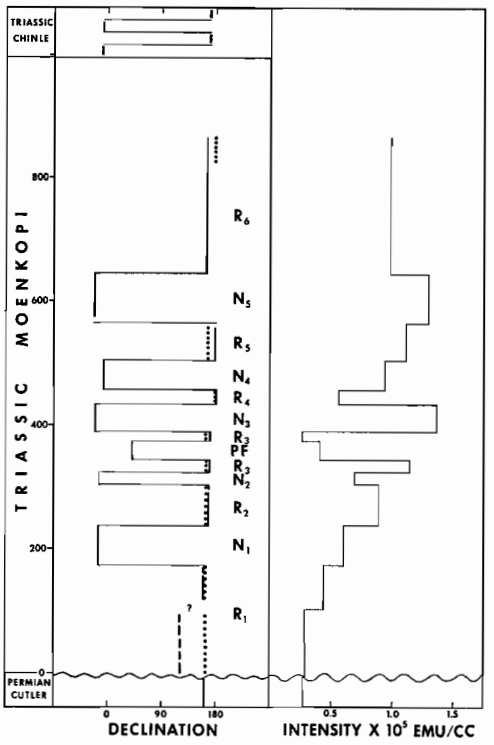


Figure 14. Variations of intensity of magnetization with stratigraphic position (right) and generalized polarity reversal sequence (left) for Lower Triassic rocks as observed in the Moenkopi Formation. Dotted lines indicate the mean declinations of the reversed intervals after 500° C thermal demagnetization. Solid lines are based on the initial NRM data.

abnormally rapid, it lends credence to the idea of a chemical origin, perhaps controlled by local ground water tables or local Eh, for much of the magnetization of the Moenkopi Formation.

POLARITY REVERSAL SEQUENCE

The detailed collections made from the lower half of the Moenkopi Formation yield the generalized polarity reversal sequence shown in Figure 14. The individual polarity zones vary in thickness from 20 to 180 feet. The uppermost interval R_6 was not sampled adequately, that is, the spacing between samples was so great that thin normal intervals could have been missed, and as a result, should not be used to imply a long period of constant reversed polarity, but rather one of primarily reversed polarity. For the lower 550 feet of the section, sampling was so dense that it is unlikely that polarity reversals were missed, that

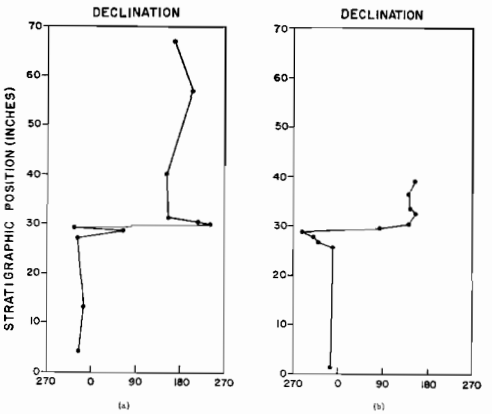


Figure 15. NRM declination changes adjacent to the $N_4 - R_5$ transition as observed in two traverses separated by about 100 yards. Note the absence of intermediate directions even when the intersample distance is only about 1 inch.

is, samples were taken at 6000- to 15,000-year intervals, if the assumption of a depositional rate of 3 cm per 1000 years is correct. Therefore, it is felt that the polarity reversal sequence from R_1 through R_5 is essentially correct, the only uncertainty being due to the possibility of a variable depositional rate. Until this same sequence is found at a nearby locality with a slightly different local geology, this remaining uncertainty cannot be removed.

The polarity and remanent directions in the lowest interval (R_1) are similar to those of Permian rocks (Irving, 1964, Helsley, 1965), the difference being less than 15°. All North American Permian results are from rocks of Lower Permian age, and if one assumes that polar wandering was uniform during the Permian, the pole position by the end of Permian time would have been almost identical to the R_1 position. Thus, the assignment of the lowest portion of the Moenkopi to the Upper Permian may be correct, and it may be possible to correlate the R_1N_1 boundary with the end of the Kiaman interval (Irving, 1964).

Heirtzler and Hayes (1967) have discussed a magnetic boundary in the North Atlantic that may correspond to the end of the Kiaman interval. If this is the correct interpretation of this boundary, the anomaly pattern seaward of it may correspond to a polarity reversal sequence for the Triassic and thus be correlatable with the Moenkopi sequence. Accordingly, the Moenkopi reversal sequence has been converted to a model sea-floor spreading anomaly pattern on the assumption that it represents 10 m.y. of time and is now at a depth of 5 km

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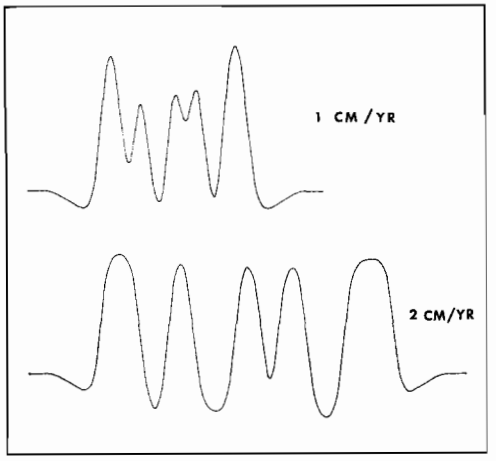


Figure 16. Magnetic anomaly patterns expected for the Lower Triassic, assuming the polarity reversal sequence represents a time of 10 m.y. PF (anomalous) interval in Figure 14 is assumed to be a local perturbation of R_3. The polarity reversal sequence upon which these curves were calculated is shown in Figure 17, location 5.

below the ocean surface. This pattern is shown in Figure 16 for spreading rates of 1 and 2 cm/year. Unfortunately, the scale of the anomaly patterns shown by Heirtzler and Hayes (1967) is so small that detailed comparisons are impossible. Nevertheless, for the anomalies just seaward of the boundary, the number of anomalies and their general character is very similar to the model anomaly pattern derived from the Moenkopi sequence. Confirmation of this tentative correlation must await measurement of the Upper Moenkopi reversal sequence, so that anomaly patterns over longer periods of time can be compared. If the correlation with the anomaly pattern at sea is correct, it suggests that the upper portion of the Moenkopi [upper Lower Triassic and Middle(?) Triassic] should be characterized by several long periods of normal and reversed polarity, rather than the relatively short intervals observed in the lower portion of the formation.

COMPARISON WITH OTHER TRIASSIC REVERSAL SEQUENCES

An attempt has been made to correlate the reversal sequence discussed above with that reported by Picard (1964) from the Chugwater Formation of Wyoming and with the results from the Buntsandstein in Europe (Burek, 1967). The sequence given by Picard is similar in that it shows thick reversed intervals and at least three thin normal zones. A tentative correlation of the Moenkopi reversals with a generalized version of the reversal stratigraphy presented by Picard is given in Figure 17. This correlation suggests that the Moenkopi section was deposited more slowly than was the Chugwater and that it represents approximately twice as much time as the section sampled by Picard. Since the section sampled by Picard ranges from the Dinwoody Formation assigned to the Lower Triassic by Kummel (1954) to the Popo Agie Formation assigned to the Upper Triassic by Colbert (1957) and sampled portion of the Moenkopi is all Lower Triassic or older, normal stratigraphic correlation would suggest that Figure 17 is in error. However, there is some evidence that the Alcova Limestone of the Chugwater Formation is of Lower Triassic age (Love, 1957; Kummel, 1954). This implies a large hiatus (or very slow deposition) between the Alcova Limestone and the Popo Agie Formation. If the Alcova is of Lower Triassic age, then the reversal correlations shown in Figure 17 are entirely consistent with the stratigraphic age assignments.

The correlation between the Moenkopi and the Chugwater is difficult for the reason that Picard reports many "intermediate reverse" and "intermediate normal" zones not found in the Moenkopi. Picard did not carry out demagnetization experiments on his Chug-

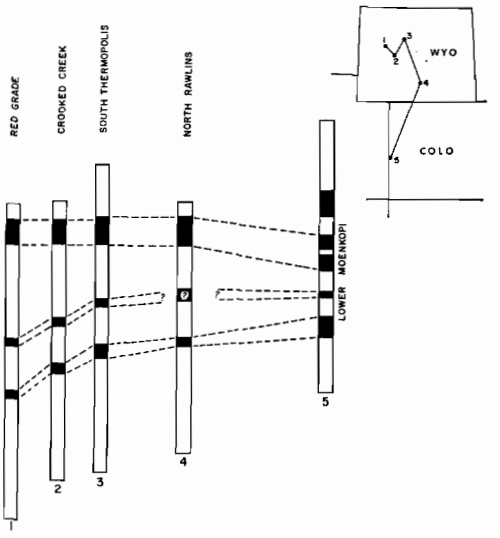


Figure 17. Proposed correlation of the Moenkopi polarity reversal sequence with the magnetic reversal stratigraphy of the Chugwater Formation of Wyoming (Picard, 1964).

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water suite, so that it is entirely possible that these "intermediate" zones are merely sections with large amounts of secondary magnetization.

The Buntsandstein data reported by Burek (1967) are also largely NRM data and show a very large clustering in the vicinity of the present pole. Thus, it is uncertain whether his normal intervals are truly normal. If the observations presented are taken at face value, they are very different from those of Picard (1964) or of this study, for they are largely of normal polarity, while those of the Chugwater and Moenkopi Formations are largely of reversed polarity. Moreover, Burek's reversal sequence starts with a long normal polarity sequence, rather than a long reversed sequence, and it has many more polarity intervals than seen by either this author or by Picard (1964). Thus, it is felt that the results presented by Burek are either not representative of the Lower Triassic polarity sequence or that they do not sample the same interval of time.

The above difficulties indicate that future studies of polarity sequences should: (1) have a high density of sampling and (2) be checked for stability even when only the polarity of the samples is at issue.

Note Added in Proof: Additional field work indicates that a mistake in correlation has been made just above the R_3 interval. In reality, the R_3-N_3 interval has been sampled twice, once in each of two neighboring sections. Therefore, Figure 2 should be corrected by removing that portion of the figure between 425 and 475 feet and substituting PF-N_3, R_4, and N_4 for PF, R_3, and N_3 in the right-hand column. This change does not appreciably change any of the conclusions reached except that it does suggest that the anomalous PF interval is at other localities a normal interval.

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